Let V be a finite dimensional representation of $\mathfrak{sl}(2)$ (the ground field in this problem is $\mathbb{C}$).

(a) Take eigenvalues of H and pick one with the biggest real part. Call it λ. Let $\bar{V}(\lambda)$ be the generalized eigenspace corresponding to λ. Show that $E|_{\bar{V}(\lambda)} = 0$.

(b) Let W be any representation of $\mathfrak{sl}(2)$ and let $w \in W$ be a nonzero vector such that $Ew = 0$. For any $k > 0$ find a polynomial $P_k(x)$ of degree k such that $E^k F^k w = P_k(H)w$. (First compute $EF^k w$; then use induction in k.)

(c) Let $v \in \bar{V}(\lambda)$ be a generalized eigenvector of H with eigenvalue λ. Show that there exists $N > 0$ such that $F^N v = 0$.

(d) Show that H is diagonalizable on $\bar{V}(\lambda)$. (Take N to be such that $F^N = 0$ on $\bar{V}(\lambda)$, and compute $E^N F^N v$, $v \in \bar{V}(\lambda)$, by (b). Use the fact that $P_k(x)$ does not have multiple roots.)

(e) Let N_v be the smallest N satisfying (c). Show that $\lambda = N_v - 1$.

(f) Show that for each $N > 0$, there exists a unique up to isomorphism irreducible representation of $\mathfrak{sl}(2)$ of dimension N. Compute the matrices E, F, H in this representation using a convenient basis. (For V finite dimensional irreducible take λ as in (a) and $v \in V(\lambda)$ an eigenvector of H. Show that $v, Fv, \ldots, F^\lambda v$ is a basis of V, and compute the matrices of the operators E, F, H in this basis.)

Denote the $(\lambda+1)$-dimensional irreducible representation from (f) by V_λ. Below you will show that any finite dimensional representation is a direct sum of V_λ.

(g) Show that the operator $C = EF + FE + H^2/2$ (the so-called **Casimir operator**) commutes with E, F, H and equals $\frac{\lambda(\lambda+2)}{2}$ Id on V_λ.

Now it is easy to prove the direct sum decomposition. Namely, assume the contrary, and let V be a reducible representation of the smallest dimension, which is not a direct sum of smaller representations.

(h) Show that C has only one eigenvalue on V, namely $\frac{\lambda(\lambda+2)}{2}$ for some nonnegative integer λ (use the fact that the generalized